



Society of Petroleum Engineers

SPE-228878-MS

Empowering Production Enhancement Success: Unveiling Malaysia's Cross Industry Insights Based on Historical Stimulation and Perforation Best Practices

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This paper was prepared for presentation at the ADIPEC held in Abu Dhabi, UAE, 3 – 6 November 2025.

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Abstract

Production Enhancement (PE) activities remain as the key contributor to Malaysia's hydrocarbon production, averaging 60kboepd annually. However, inconsistent success rates have led to value leakage. To address this, Malaysia Petroleum Management (MPM) conducted a comprehensive five-year lookback study across four major operators. This paper presents critical insights, data-backed findings, and actionable recommendations to improve high impact PE activities specifically stimulation and perforation, ensuring enhanced success rates, cost efficiency, and business sustainability in Malaysia's fields.

A centralized Power BI database was developed, consolidating over 200 PE job records from 20+ fields across the three regions, covering both shallow and deepwater assets. The study focused on stimulation and perforation, two of the most frequently executed yet consistently value-leaking well intervention types, where high activity levels have not translated into proportional business gains. Data was sourced from well models, well completions, well intervention reports, contractor proposals and pre/post-job analyses. Jobs were classified as "success" or "failure" based on production gains and economic returns. Forensic analysis of failed jobs identified recurring gaps in candidate selection, treatment design, fluid system application, and execution practices. The study provided operators with visibility on technical and operational pitfalls, establishing cross-operator benchmarks and distilling best practices into practical recommendations aimed at enhancing future PE proposals and execution strategies. The lookback analysis revealed that perforation jobs had 50% failure rate, with cement packer additional perforations being the costliest and least effective. Stimulation jobs showed only 54% success rate, despite notable production gains. Retarded acid treatments consistently outperformed than other stimulation fluid systems. Post-study implementation of recommendations resulted in tangible improvements: perforation success rates increased from 57% to 80%, while stimulation success rates soared from 54% to 83%. These enhancements translated into substantial cash generation and improved field economics. Key learnings include addressing reservoir complexities, managing post-job water/gas production, optimizing perforation strategies, and refining damage diagnosis and treatment fluid selection.

While there is a significant abundance of case histories available for PE jobs, the industry notably lacks comprehensive look-back studies based on substantial field data from multiple operators across deepwater and shallow water assets. This paper aims to bridge this gap by offering Malaysia operators valuable knowledge and best practices for attaining success in realizing promised gain from high-value impact PE activities, stimulation and perforation.

Introduction: Positioning PE/IWR as Strategic Lever for Malaysia's Oil & Gas Production Sustainability

In recent few years, operators in Malaysia have intensified activities to revive idle wells and enhance production/injection from active wells. This was driven by the requirement to sustain base production, volatile oil price, aftermath of the COVID-19 pandemic situation and availability of resources. Depleted reservoir pressure, marginal fields, higher production deferment, aging facilities and well integrity issues only add to the complications. Production Enhancement (PE) and Idle Well Reactivation (IWR) have emerged as essential strategic levers in safeguarding Malaysia's oil and gas industry. However, the overall success rate of each activity and efficiency of the operation was not encouraging within 50-60% only and no conclusive benchmarking exercise on the past years available to compare the overall performance across the operators in Malaysia.

Figure 1 outlines that between 2021 and 2024, Malaysia's PE/IWR workstream delivered substantial values that generating an estimated RM15.35 billion (annualized) in cumulative cash flow against an incurred expenditure of RM6.28 billion (exchange rate equivalent RM4.50~USD1). This nearly three times return on investment underscores the continued commercial viability of PE/IWR activities in extending field life and sustaining national production. However, the operating environment is becoming increasingly complex, marked by rising service costs, resource constraints, and dwindling inventory of technically straightforward opportunities. This necessitates a more rigorous and data-driven approach in evaluating, prioritizing, and executing future interventions to sustain economic performance as explained by Hasmizah et al (2022).

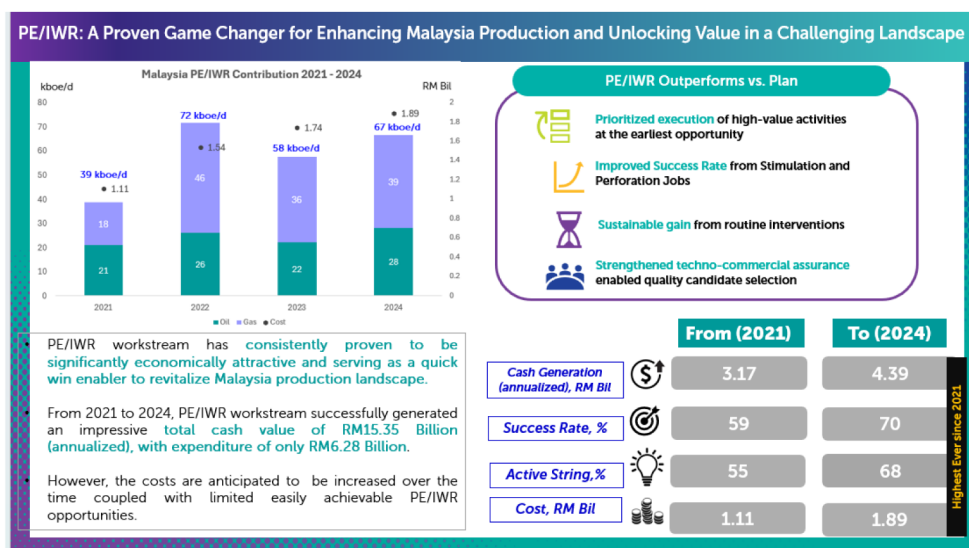


Figure 1—PE/IWR Performance and Contributions to Malaysia Production (Year 2021 till 2024)

In response to these evolving challenges, as the host authority of Malaysia hydrocarbon resources, MPM has commissioned a structured, five-year cross-industry performance study covering over 200 PE jobs

conducted across shallow and deepwater assets by four major operators from 2021 to 2023. The initiative was structured in two sequential phases:

Phase 1: PE IWR Performance Lookback Analysis. Phase 1 was initiated to evaluate historical performance across Malaysia by assessing production gain, Unit Enhancement Cost (UEC) and identifying value-creating versus value-eroding interventions. It provides critical insights into performance trends, recurring technical failures, and underperforming activities, while also highlighting replicable best practices. These statistics enable data-driven decision-making, benchmarking across operators, and more accurate forecasting for future planning.

Stimulation and Perforation jobs emerged as key underperforming categories, characterized by high implementation frequency but inconsistent outcomes and significant value leakage. The findings from Phase 1 signalled the need for enhanced technical due diligence and cross-operator standardization.

Phase 2: PE IWR Deep-Dive Performance Analysis. The deep-dive analysis on stimulation and perforation focuses on addressing key technical and operational gaps to enhance job success rates and value creation. For stimulation, the emphasis is on refining candidate selection through better reservoir diagnostics, ensuring more effective treatment designs tailored to formation characteristics, and improving the accuracy of production gain forecasts. In parallel, the perforation analysis aims to strengthen candidate screening processes, optimize gun system selection for targeted intervals, and implement benchmarking of results across regions and operators to identify performance drivers. Both interventions prioritize cost efficiency through better upfront planning and job execution, ultimately aiming to minimize value leakage, reduce rework, and improve return on investment. These improvements are essential for sustaining the commercial viability of PE/IWR campaigns, particularly in Malaysia's aging fields where margin and resources are increasingly constrained.

Systematic and Data Driven Framework for Performance Lookback

This section outlines the systematic approach used to conduct Lookback Analysis of stimulation and additional perforation activities through a centralized Microsoft Power BI database. The methodology details the end-to-end workflow from data gathering to final performance evaluation as explained by Figure 2. By applying a systematic and data-driven framework, the analysis ensures consistency, transparency and meaningful interpretation of the results. The following outlines the key steps and processes involved in the project:

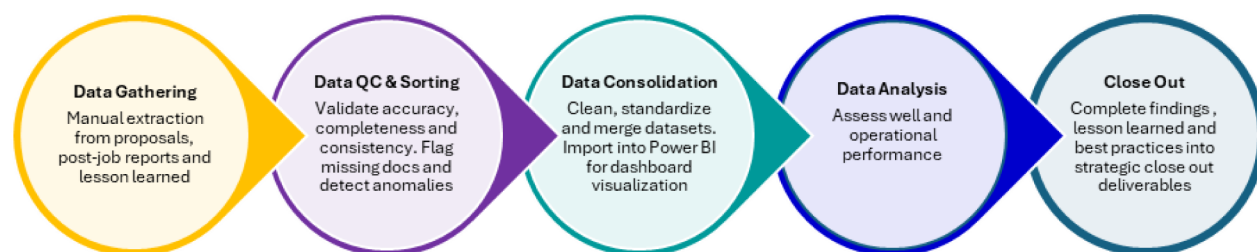


Figure 2—Performance Lookback Workflow

Step 1 – Data Gathering

The first and most critical stage involves manual extraction of information from multiple technical and operational sources. These include Techno-Com Proposals, Post Job Reports from contractors, and Post Job Analysis & Lesson Learnt Reports. Each document provides unique insights from design intent to execution details and observed outcomes. All extracted information is systematically organized by Well ID, enabling accurate tracking of each well's performance history.

Step 2 – Data Quality Check & Sorting

Once data is gathered, it undergoes a thorough validation process to ensure completeness and consistency. Each well is checked for the availability of the three critical documents; Techno-Com Proposal, Contractor Post Job Report and Post Job Analysis & Lessons Learnt Report. Wells missing any of these are flagged as Not Available (N/A), indicating the need for follow-up or exclusion from the analysis. This phase also includes anomaly detection where irregular or questionable data points are flagged for further verification.

Step 3 – Data Consolidation

Validated data is then cleaned and merged into a single master dataset. The consolidated dataset is imported into a Power BI Data Checklist Dashboard as shown in Figure 3, which visually displays the status of document availability (Yes/ N/A) for each well. This visualization allows quick filtering, rapid identification of data gaps and prioritization of wells for further analysis.

Job ID	Category	Contractor	Job Date	SWAC Proposal	PIA/LLP Submission	PR Contractor	Job Details
Well 1	Acid Solution	Dimension Bid	Thursday, 28 May, 2020	Yes	Yes	Yes	Scale and Wax Treatment
Well 10	Non-Acidic Solution	SVP	Friday, 24 May, 2019	Yes	Yes	Yes	Organic & inorganic removal treatment
Well 11	Non-Acidic Solution	SVP	Thursday, 18 July, 2019	Yes	Yes	Yes	Acid Simulation - To remove near wellbore damage suspect due to fines migration
Well 12	Non-Acidic Solution	SVP	Friday, 24 May, 2019	Yes	Yes	Yes	Acid Simulation - Unsuccessful operation for wax and fines issue
Well 13	Acid Solution	Dimension Bid	Wednesday, 15 April, 2020	N/A	N/A	Yes	Acid Simulation - To remove near wellbore damage caused by lost circulation material pumped during workover
Well 14	Acid Solution	Dimension Bid	Friday, 1 May, 2020	N/A	N/A	Yes	Scale Removal Treatment
Well 15	Non-Acidic Solution	Dimension Bid	Monday, 28 December, 2020	N/A	N/A	Yes	Acid Simulation - To remove the fines plugging the gravel pack
Well 16	Non-Acidic Solution	Dimension Bid	Wednesday, 1 January, 2020	N/A	N/A	Yes	Acid Simulation - To remove the fines plugging the gravel pack
Well 17	Non-Acidic Solution	Dimension Bid	Wednesday, 1 January, 2020	N/A	N/A	Yes	Acid Simulation - To remove the fines plugging the gravel pack
Well 18	Non-Acidic Solution	Dimension Bid	Tuesday, 14 January, 2020	N/A	N/A	Yes	Acid Simulation - To remove the fines plugging the gravel pack
Well 19	Non-Acidic Solution	SVP	Thursday, 26 September, 2019	Yes	N/A	Yes	Acid Simulation - Post TTSP installation for fines migration
Well 2	Non-Acidic Solution	SVP	Monday, 21 October, 2019	Yes	N/A	Yes	Acid Simulation - To reduce skin post TTSP installation and fines plugging
Well 20	Non-Acidic Solution	SVP	Saturday, 14 November, 2020	Yes	N/A	Yes	Organic removal treatment (R&A)
Well 21	Acid Solution	Dimension Bid	Friday, 11 June, 2021	N/A	N/A	Yes	Acid Screen Wash - Remove Scale Build up
Well 22	Acid Solution	Dimension Bid	Friday, 25 June, 2021	N/A	N/A	Yes	Acid Simulation to reduce skin across CHS&S and near wellbore region, remove sulfate scale
Well 23	Acid Solution	Dimension Bid	Monday, 14 June, 2021	N/A	N/A	Yes	Scale Removal Treatment & Acid Simulation
Well 24	Acid Solution	SVP	Saturday, 22 October, 2022	Yes	N/A	Yes	Acid Simulation - Remove near wellbore damage due to fines migration
Well 25	Acid Solution	SVP	Sunday, 24 July, 2022	Yes	N/A	Yes	Acid Simulation - Perform acidizing using 55 Acid to reduce skin
Well 26	Acid Solution	SVP	Tuesday, 9 August, 2022	Yes	N/A	Yes	Acid Simulation - Perform acidizing using 55 Acid to reduce skin
Well 27	Acid Solution	SVP	Monday, 25 July, 2022	Yes	N/A	Yes	Acid Simulation - Perform acidizing using 55 Acid to reduce skin
Well 28	Acid Solution	SVP	Friday, 8 June, 2018	Yes	N/A	Yes	Acid Simulation - Remove near wellbore damage due to mud losses and fines migration
Well 29	Acid Solution	SVP	Wednesday, 27 June, 2018	Yes	N/A	Yes	Acid Simulation - Remove near wellbore damage due to fines migration
Well 3	Non-Acidic Solution	SVP	Saturday, 5 October, 2019	Yes	N/A	Yes	Acid Simulation - To reduce skin in IGP and formation due to fines migration and plugging
Well 30	Acid Solution	SVP	Saturday, 23 June, 2018	Yes	N/A	Yes	Acid Simulation - Remove near wellbore damage due to fines migration
Well 31	Non-Acidic Solution	SVP	Monday, 30 July, 2018	Yes	N/A	Yes	Acid Simulation - Remove near wellbore damage due to fines migration

Figure 3—Example of Data Check List

Step 4 – Data Analysis

The prepared dataset is analysed to evaluate both well performance and operational performance. Key performance metrics such as production gain, Productivity Index (PI) changes and sustainability of gains are assessed to determine job success or failure. Interactive dashboards are used to present the results, allowing users to view overall statistics and drill down into individual performance detail. Both stimulation and add perforation activities are analysed systematically with results compared against success criteria.

Step 5 – Close-out

The final stage compiles all findings, lessons learned and best practices into a comprehensive close-out package. This includes the cleaned dataset, Power BI dashboards, performance summaries and recommendations for improving future stimulation and perforation interventions. Insights from unsuccessful jobs are translated into practical guidelines to enhance job design, execution and monitoring. The close-out process ensures that the Lookback Analysis is not merely a historical review but a fronting tool that feeds into continuous improvement and optimizes production.

Data Source for Lookback and Deep Dive Analysis

The Lookback Analysis integrates on multiple data sources to ensure a comprehensive and reliable evaluation of both acid stimulation and add perforation jobs. Four primary sources were utilized: well models, well completions, contractor proposals, and post job reports. Each of these sources contributes unique technical insights essential for performance benchmarking and root cause analysis. Well models

provide critical reservoir and production characteristics including permeability, porosity, pressure and skin which form the basis for understanding pre-job conditions and predicting post job behaviour. Well completion diagrams offer detailed mechanical configurations such as perforation intervals, tubing sizes, and completion type which are important for validating job design coverage and evaluating operational constraints that may influence treatment outcomes. Contractor proposals include the planned job design, chemical volumes, deployment methods, and stimulation or perforation schedules. These documents serve as the baseline for assessing the alignment between planned and executed operations. Finally, post-job reports provide actual field execution details including volumes pumped, pressure responses, job duration, operational issues encountered and remarks on treatment effectiveness. Together, these data sources were quality-checked and cross-validated before being integrated into the central Power BI database ensuring consistency across the analysis.

Centralized Power BI Tool for Performance Tracking and Lessons Learned

Power BI central database was developed to provide a structured and comprehensive Lookback analysis. The core components of the Power BI Database consist of:

1. Data Checklist

- Serves as the foundation for data completeness and accuracy.
- Tracks essential information such as well identifiers, job type, chemical volumes, gun details, and pre/post-job production rates.
- Flags missing or inconsistent data, ensuring only reliable records are analysed.

2. Data Overview

- Provides a high-level summary of all jobs analysed.
- Breaks down jobs by year, type (acid stimulation or add perforation), status, treatment category and operator.
- Highlights trends, coverage gaps and overall activity levels over time.

3. Pre-Job vs Post-Job Performance

- Compares well performance before and after the intervention.
- For stimulation; evaluates production rate (liquid and oil), water cut, skin changes and segmented by chemical type.
- For additional perforation; measures production gains, cost impacts and gun performance.

4. Operation Summary

- Summarizes execution details for add perforation jobs, including deployment methods and job status per category.
- Enables benchmarking against previous operations to refine execution strategies.

5. Failure Classification

- Tags each job as "success" or "failure" against predefined performance criteria.
- Supports statistical breakdown by treatment type and execution method.

6. Root Cause Failure Analysis

- Investigates why certain jobs underperformed.

- Links operational parameters, well conditions and treatment designs to common failure causes such as poor candidate selection, design uncertainties or execution deviations.

7. Lessons Learned

- Captures insights from both successful and fail jobs.
- Provides actionable recommendations to improve job design, execution quality and operational safeguards.

The Power BI dashboard serves as a powerful tool for operators to efficiently monitor job performance, identify success or failure trends and analyse the root causes behind each outcome. Figure 4 illustrates Power BI dashboard developed for the lookback analysis of both Additional Perforation and Stimulation jobs. The dashboard encompasses comprehensive sections, including structured data checklists, in-depth analytical visualizations, conclusive findings and consolidated best practice recommendations.

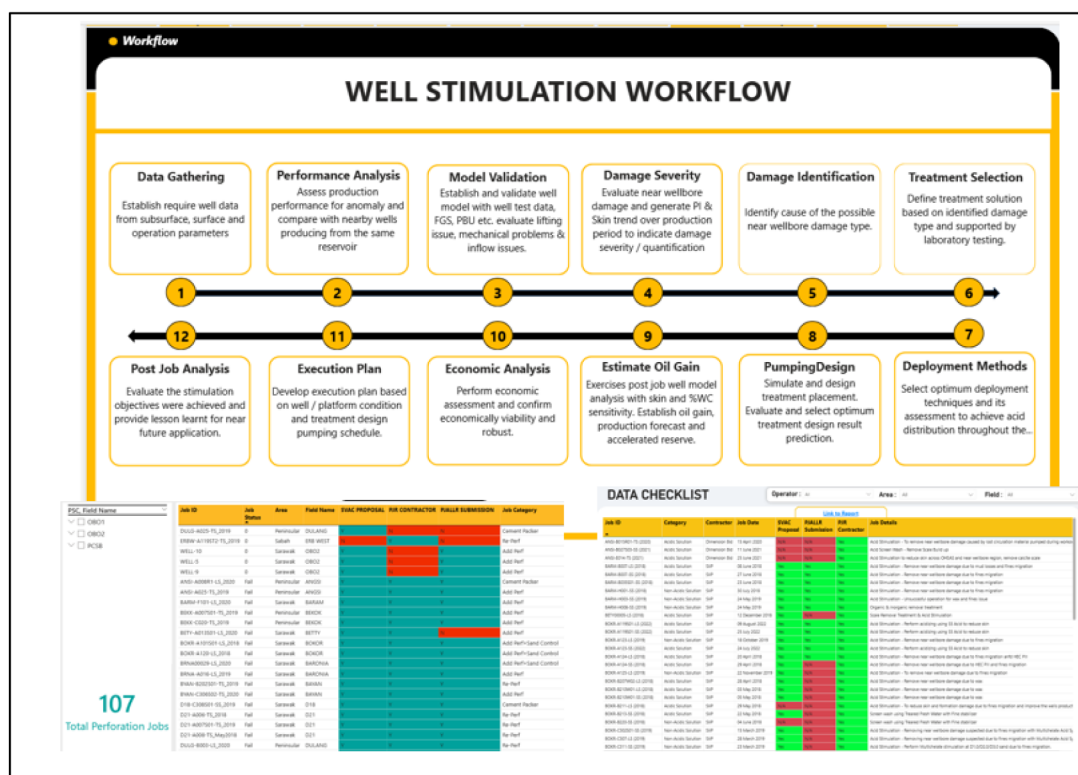


Figure 4—Power BI Dashboard Additional Perforation and Stimulation Analysis

This dashboard facilitates the systematic identification of repeated operational trends in both Add Perforation and Stimulation jobs. The dashboard accelerates the replication of effective execution strategies while providing a proactive framework for mitigating potential value losses and suboptimal performance in future interventions.

Standardised Criteria for Post Job Outcome Classification

In this study, each job was assessed using standard performance metrics to determine whether it was classified as successful or fail. The primary evaluation method involved comparing pre-job and post-job production data including liquid rate, oil rate, water cut percentage, and skin (for stimulation) to quantify the actual impact of the intervention. A job was classified as a success if post-job performance demonstrated sustained improvement above the defined threshold. The cost perspective was also considered in classifying

the jobs for root cause analysis. By analysing the distribution and variance between planned and actual job costs, further insights were obtained into influencing factors such as job category, deployment method and operational execution. This allowed deeper root cause analysis, ensuring lessons learned are anchored in both technical and economic perspectives. Figure 5 presents a plot of planned versus actual gains and costs for additional perforation jobs. Strings in the green-shaded zone are generally classified as successful while those in the red-shaded zone are considered unsuccessful.



Figure 5—Plot Plan against Actual Data for Additional Perforation

In-Depth Review Process for Identifying Root Causes

As previously discussed, the job outcome is initially classified based on post-job production performance versus the planned gain. Once the classification is established, a root cause analysis is conducted to determine the factors that contributed to the result. This analysis covers but is not limited to the following aspects:

- **Completion Design & Execution:** For additional perforation jobs, a detailed review is conducted on the perforation gun type, charge performance, gun depth placement, and operational execution (including gun firing confirmation and underbalance conditions).
- **Chemical Effectiveness:** For stimulation jobs, the type of acid or chemical package, quality of additives, design parameters (such as volume, rate, and contact time), and diversion techniques (if applied) are evaluated.
- **Reservoir Compatibility:** The formation type, mineralogy, skin factor and near-wellbore damage mechanism are examined to assess how well the chemical treatment matches reservoir conditions.
- **Post-Job Surveillance & Diagnostics:** Post-intervention data including pressure trends, changes in fluid composition, well test results, and production logs (if available) are analysed to verify the impact of the job.
- **Economic Review:** Even when technically successful, a job may be deemed economically unattractive if costs significantly exceed the expected Unit Enhancement Cost (UEC) and/ or if the payback period is too long.

By integrating production data, job design parameters and field observations, the in-depth review process enables the identification of recurring patterns of success and helps avoid repeating past mistakes.

Deep Dive Analysis for Historical Stimulation

This section presents the key findings from a comprehensive deep dive analysis of 63 stimulation jobs conducted by four major operators in Malaysia. Figure 6 displays the overview of chemical

OVERVIEW

Operator:
Area:
Field:

Operator

Operator	Count	Percentage
OBO	10	16%
PCSB	58	84%

State

State	Count	Percentage
Sarawak	11	17%
Sabah	13	13%
Perlis	27	27%

Status

Status	Count	Percentage
Success	27	43%
Fail	7	21%
Suspended	24	34%

Job Status By Operator

Operator	Fail	Success	Suspended
PCSB	10	48	10
OBO	0	10	0

Main Damage Type

Damage Type	Count	Percentage
Free Migration	6	17%
Intra Ovarian	7	17%
Organic desorption (9%)	1	1%
Organic desorption (Emulsion)	49	73%
Dyeing/Carpetation/Waterborne...	27	27%

Job Status By Year

Year	Fail	Success	Suspended
2018	10	48	10
2019	10	48	10
2020	10	48	10
2021	10	48	10
2022	10	48	10

Treatment Category

Treatment Category	Count	Percentage
Acidic Solution	27	42.89%
Non-Acidic Solution	35	57.14%

63 Acid Stimulations Job

Operator	Job ID	Reservoir	Status
PCSB	ANG-8015021-T5	(2020)	Success
PCSB	ANG-80275203-S5	(2021)	Success
PCSB	ANG-4204-T5 (2021)		Success
PCSB	BARM-8007-L3 (2018)		Fail
PCSB	BARM-8007-S5 (2018)		Fail
PCSB	BARM-8025501-S5	(2018)	Fail
PCSB	BARM-0001-S5 (2018)		Success
PCSB	BARM-0001-S5 (2018)		Success
PCSB	BARM-0008-S5 (2018)		Suspended
PCSB	BARM-0008-S5 (2018)		Fail
PCSB	98710020-L3 (2018)		Fail
PCSB	BOCH-4119501-L5, L7, L8, L15, L31, L60 (2020)		Success
PCSB	BOCH-4119501-L5, L7, L8, L15, L31, L60 (2020)		Success

The key findings from the lookback study related to the stimulation activity were as follow:

- Downloaded from <https://onepetro.org/SPEADIP/25ADIP/conference/25ADIP> by Dmitry Kruglov on 08 November 2025

inaccurate candidate selection contributed the least, at 5%, indicated that while most wells were screened appropriately, a small portion of candidates with low stimulation potential were still selected.

2. Design and operational issues were identified as the leading causes of stimulation failure. A major contributor was the oversight of tubing pickling prior to pumping the main treatment fluids which reported for 71% of failures within this category as shown in [Figure 9](#). Skipping this critical step allowed iron compounds and deposits to enter the formation leading to unintended damage. Bullheading was the preferred deployment method over coiled tubing units (CTU) due to cost-effectiveness and smaller operational footprint; however, the absence of tubing pickling during bullheading operations significantly increased the likelihood of treatment failure.
3. Sub-optimal chemical selection was identified as one of significant contributor to stimulation failures. A common example was the treatment of fines migration with non-acidic systems. Since fines migration is typically associated with siliceous material, effective removal requires acid-based systems such as retarded acid or mud acid. Inappropriate fluid selection, particularly the use of non-acidic systems, was frequently observed in field applications. The mismatch between damage mechanism and treatment fluid led to poor stimulation effectiveness. Instead of dissolving or mobilizing fines, non-acidic fluids often displaced particles deeper into the formation, created additional blockage and further impairing permeability. Analysis of treatment data [Figure 10](#) demonstrated that such mismatches were a key reason behind reduced success rates. Incorporating pre-flush solvents or pre-flush acids was shown to improve treatment performance particularly for candidates with combined damage mechanisms (e.g., fines migration coupled with organic or inorganic deposition). Treatments that included a pre-flush solvent demonstrated higher success rates, though failures could still result from other operational or reservoir-related parameters. Laboratory testing is therefore essential prior to field application as it provides critical insight into the chemical interactions between deposition materials, formation fluids and treatment fluids. This ensures that the selected chemical system is tailored to the actual damage type, thereby reducing the risk of failure and enhancing stimulation success, shown in [Figure 11](#).
4. Inaccurate formation damage identification accounted for 19% of stimulation failures. As shown in [Figure 12](#), these cases were commonly associated with incomplete diagnostic evaluations and challenges coming from through-tubing perforations and remedial sand control operations, which cannot be effectively addressed using acid systems.

A lack of detailed understanding of reservoir mineralogy also contributed to misdiagnosis. In Malaysian sandstone reservoirs, fines migration is the dominant damage mechanism, as demonstrated in [Figure 15](#) due to relatively high clay content (>5%) particularly illite and kaolinite. Misidentifying the actual damage mechanism led to the selection of ineffective treatment fluids and designs. For example, treating perforation-related damage or sand control issues with acid systems resulted in little to no production gain. In addition, fines migration often occurred in combination with other damage types such as organic (e.g., wax) and inorganic deposition, further complicating treatment outcomes. Failure to justify for these combined mechanisms resulted in sub-optimal chemical selection and incomplete damage removal. A more comprehensive diagnostic workflow is essential to reduce the risk of misdiagnosis. Laboratory analyses such as produced water chemistry, SARA (Saturates, Aromatics, Resins, and Asphaltenes) and mineralogical characterization using XRD/XRF provide critical insights into deposition materials and secondary damage mechanisms. Incorporating these diagnostics enables more accurate identification of the dominant and secondary damage types, ensuring that chemical treatments are tailored to the actual reservoir conditions.

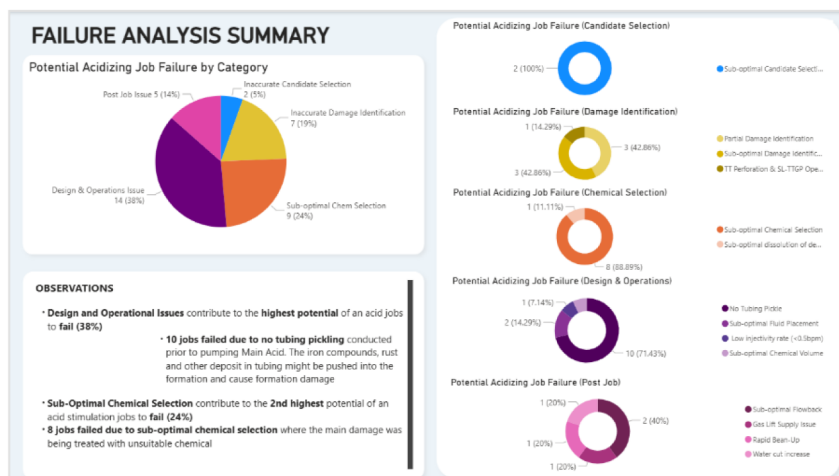


Figure 8—Historical Stimulation Failure Classification and Analysis Summary

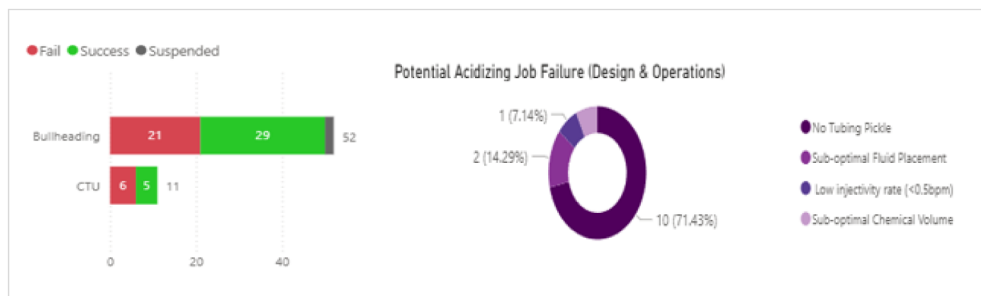


Figure 9—Deployment method for stimulation activity (left) and summary of design and operations issue failure (right)

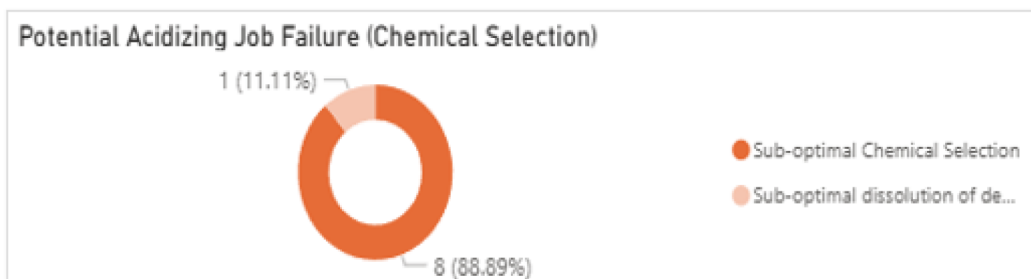


Figure 10—Summary of Chemical Selection Failure

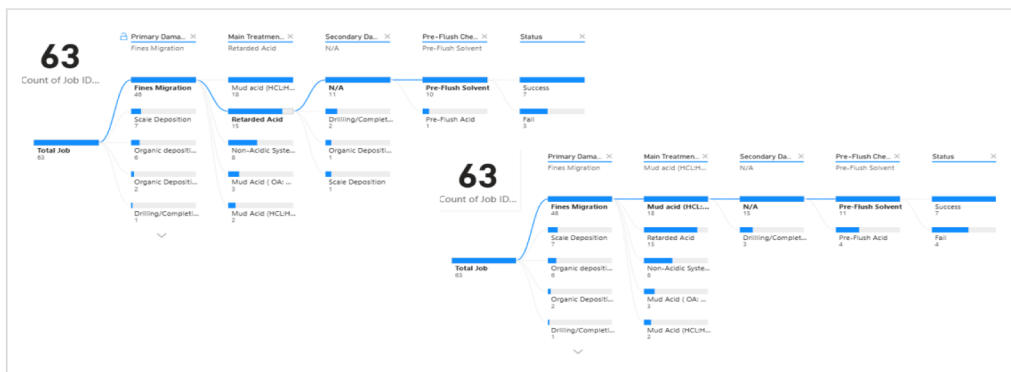


Figure 11—Summary of Root Cause Analysis for Failure Stimulation due to Chemical Selection

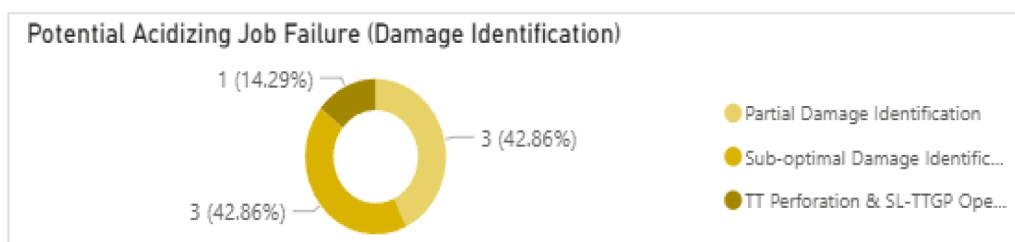


Figure 12—Summary of Chemical Selection Failure

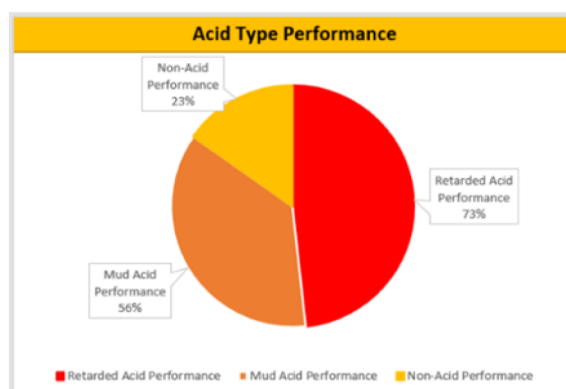


Figure 15—Stimulation Treatment Fluids Performance

Stimulation Treatment Fluids System

The selection of chemical system was a key factor in stimulation performance. In many cases, treatments did not align with the actual damage mechanisms and/or were not effective in addressing complex near-wellbore conditions. Convectonal mud acid system (hydrochloric acid, HCl/hydrofluoric acid HF) was often challenged by fluid compatibility issues and risk of secondary or tertiary precipitation, while non-acidic systems were largely ineffective against fines migration and other prevalent damage types in Malaysia sandstones. Figure 15 shows that retarded acid treatments achieved the highest success rate (73%) as compared to mud acid and non-acidic systems. Retarded acid not only delivered better skin reduction and sustainable improvements in oil and liquid rates but also minimized water cut increase compared to other systems and it is documented by Ting, C. S. et al (2019). In contrast, mud acid treatments gave only moderate production gains with higher water cut, while non-acidic systems often failed to remove the primary damage mechanism and in some cases, worsened water production.

Treatment Fluids Performance on Skin Effect Post Stimulation

Skin reduction was one of the key parameters monitored following stimulation, as it directly reflects the effectiveness of the treatment. Post-job reports showed that both retarded acid and mud acid achieved significant skin reduction which translated into higher PI and improved liquid and oil rates. This outcome reflects the successful removal of fines migration damage, primarily alumino-silicates materials that block the pore throats as explained in Mohd Jeffry, S. J (2020). Mud acid treatments typically achieved greater skin reduction because of their strong reactivity. The presence of HCl in mud acid enhances its ability to dissolve clays, feldspar, quartz fines, carbonate material and iron oxides. In contrast, retarded acid exhibited a slower reaction rate, as its delayed HF release targets clays and silicates specifically, allowing for deeper acid penetration and better controlled dissolution within the formation as mentioned by Mohd Jeffry, S. J (2020). On the other hand, non-acidic systems showed minimal impact on skin reduction, as the dominant damage type could not be dissolved by these fluids. This limitation was also reflected in the insignificant increases observed in liquid and oil production when non-acidic systems were applied. Non-acidic systems are typically applied in cases of wells experienced combined damage, such as wax deposition and fines

migration, which partly explains their limited effectiveness. The skin effect trend by different type of stimulation fluids is displayed in Figure 16.

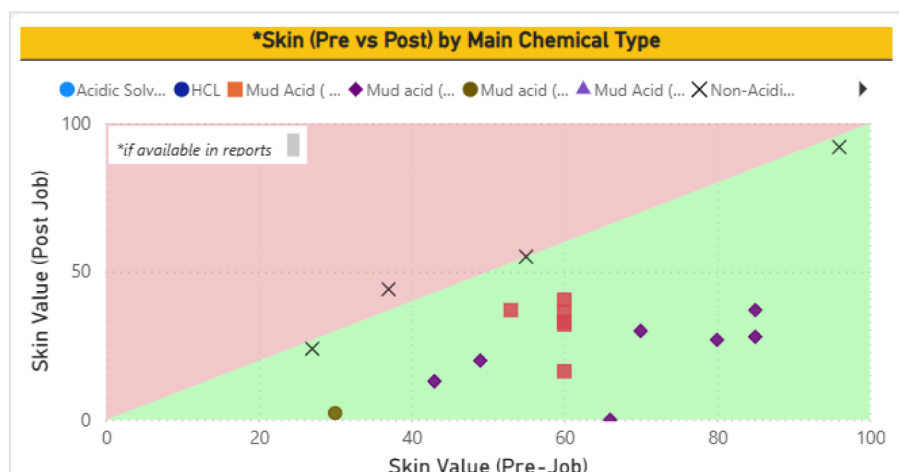


Figure 16—Skin Effect Pre and Post Stimulation based on Different Treatment Fluids Application

Treatment Fluids Performance on Water Production Post Stimulation

Incremental water production after stimulation was primarily influenced by the treatment fluids deployment method and reservoir conditions. Bullheading, the most common technique, often caused uncontrolled acid distribution, increasing the risk of fluid entry into high-permeability water-bearing zones. As noted by Mohd Jeffry, S.J. (2020) and Economides, this challenge is more noticeable in brownfield operations with prevalent water-bearing intervals. The lack of diversion techniques, such as chemical diverters, further increased the risk of acid channelling into water zones. Additionally, stimulation performed close to the oil–water contact (OWC) can enhance vertical permeability and promote water coning.

Most chemical treatments resulted in higher water cut post-stimulation as showed in Figure 10, particularly with mud acid due to its aggressive reactivity and broad mineral dissolution capacity. As a result, some treatments unintentionally increased water production along with oil. However, analysis also revealed some cases where retarded acid systems reduced water cut. This occurred because retarded acid selectively improved oil productivity by penetrating tighter oil-rich intervals while avoiding extensive stimulation of water-bearing zones, thereby lowering the overall water cut.

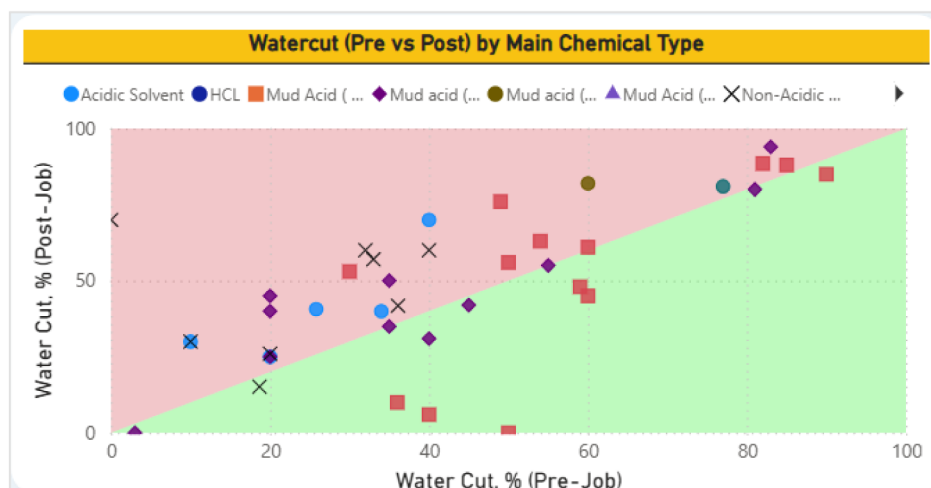


Figure 17—Water Cut (%) Pre and Post Stimulation based on Different Treatment Fluids Application

To minimize post-stimulation water cut increases, deployment methods must be optimized. The use of diversion techniques, particularly chemical diverters, can improve fluid placement and reduce acid entry into water-bearing zones. Extensive reservoir characterization, including detailed understanding of the OWC and zonal heterogeneity, is critical for treatment planning. Retarded acid systems, with their delayed reactivity, demonstrated better selectivity in bypassing high-permeability water zones and targeting tighter oil-bearing intervals.

Stimulation Best Practices and Learnings

Successful acid stimulation cases typically share common benchmarks across three critical aspects:

- *Treatment Recipe*: The most effective treatments come from multi-stage fluid systems incorporating retarded acids or mud acid variants, which provided better control over acid-rock interaction. For example, aluminium-complexed retarded mud acids work well in sandstone because they react slower with siliceous minerals, allowing the acid to penetrate deeper and reduce secondary damage Ji (2017). Successful treatments also include additives in the pre-flush stage to remove organic residues or fines and ammonium chloride in the post-flush stage to prevent harmful precipitates. Successful cases also included secondary damage treatment additives in the pre-flush stage, enhancing removal of organic residues or fines while ammonium chloride (NH₄Cl) was included in the post-flush phase to prevent precipitation.
- *Placement Techniques*: Operational success was strongly associated with tubing pickling to eliminate scale or rust that could obstruct acid flow and the use of diversion methods (e.g., ball sealers or chemical diverters) to achieve uniform acid distribution across the targeted intervals. Additionally, control over injection volumes and pump rates was vital to ensure that the acid contacted the intended damaged zones without premature breakthrough or channelling.
- *Clean-Up & Flowback*: The clean-up and flowback stage was equally critical. Jobs using HF based systems showed improved results when immediate flowback was executed to remove reaction of byproducts such as fluorosilicates, which can form damaging precipitates if left in the formation. In scenarios where flowback was delayed, the use of over flush fluids helped displace residual acid and reduce the risk of emulsion or sludge formation. Furthermore, coordinated multiple stimulation in a campaign improved logistical efficiency, reduced downtime, and enhanced overall stimulation success rate.

Summary of Recommendations from The Deep Dive Analysis for Historical Stimulation

The lookback analysis of stimulation campaigns has revealed recurring technical and operational gaps that contributed to sub-optimal success rates. To achieve the targeted success rate for stimulations, the following recommendations are proposed:

1. Candidate Selection and Pre-Job Screening
 - Reassess water cut (WC) prior to execution to confirm gain sustainability and economic viability. High water cut wells should be replaced with more promising candidates (WC <70%) to avoid undesired outcomes Mohd Jeffry et al (2020)
 - Promote multidisciplinary collaboration in proposal preparation to ensure knowledge transfer and comprehensive technical evaluation.
 - Exclude wells with irreversible formation damage from past interventions (e.g., excessive HEC pill usage, through-tubing gravel packs) as such conditions are unlikely to be restored through acid stimulation.
2. Damage Diagnosis and Reservoir Characterization

- Prioritize comprehensive laboratory testing tailored to the specific reservoir, including produced water analysis, SARA, XRD/XRF, sludge tendency and scale/wax sample characterization.
 - Avoid reliance on generic test data from other fields; field-specific testing is essential for accurate damage identification.
 - Develop reservoir damage mapping across Malaysian assets, documenting mineralogy, scale types and organic deposition tendencies to support future stimulation planning.
3. Chemical Selection and Database Development
- Establish a centralized database of stimulation chemicals with documented advantages, limitations, laboratory results and field performance.
 - Conduct rigorous laboratory testing (dissolution, compatibility, core flooding, corrosion) to ensure chemical formulations are effective for the identified damage mechanisms.
4. Treatment Design and Execution
- Implement proven design practices, including tubing pickling, fluid sequencing (pre-flush, main acid, post-flush) and contingency over-flushes, which have shown improved treatment outcomes.
 - For bullheading operations in intervals exceeding 30 ft, incorporate water diverters to improve fluid placement efficiency.
 - Execute stimulation jobs in campaign mode to optimize costs and logistics.
5. Operational Reliability and Post-Treatment Practices
- Ensure proper maintenance of stimulation and testing equipment, both on platform and vendor-supplied, to enable reliable flowback and accurate post-job evaluation.
 - Adhere to established bean-up procedures, particularly for wells susceptible to fines migration.
 - Secure adequate gas lift availability to support well cleanup and sustained post-treatment production.
 - Conduct continuous post-job monitoring to assess production sustainability and treatment effectiveness.

Deep Dive Analysis for Historical Additional Perforation

As mentioned in previous sections, perforation is a crucial PE/IWR activity for the operators considering most assets are in "brown fields" category. Other non-routine interventions to boost production are generally more costly, making perforation a more favourable option. In this lookback study, near to 100 through tubing perforation jobs were studied in detail. The perforation activity recorded a success rate of ~57%. The lookback study was therefore undertaken to identify gaps and explore opportunities to improve the success rate towards 80%. [Figure 18](#) provides a breakdown of the jobs into success, failure and suspended categories. It also highlights that most jobs delivered lower actual gains than the planned gains. The key findings, failure causes and learnings will be detailed in rest of the section.

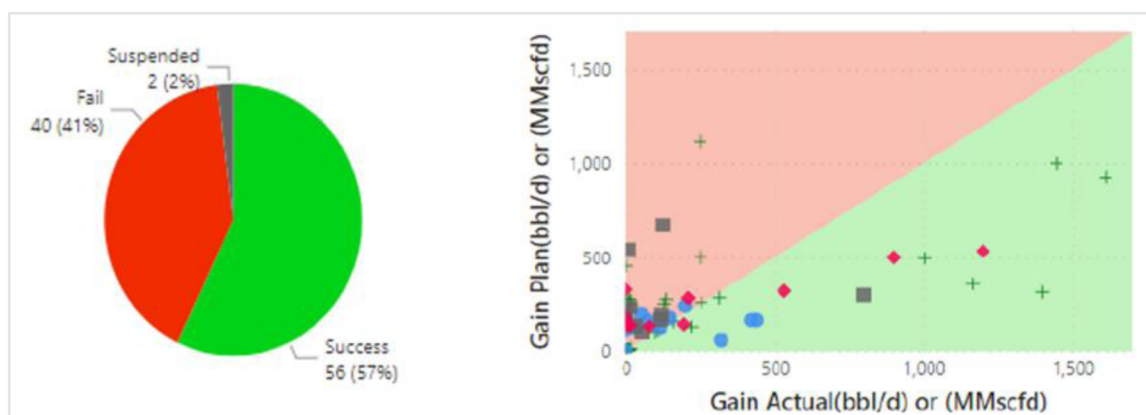


Figure 18—Low Success Rate for Through Tubing Perforation (left) and Lower than Planned Gain for Majority of the Perforation (right)

The key findings from the lookback study related to the through tubing perforation activity were as follows:

1. The key reasons for the failures from the lookback study could be distributed into following categories: (a) High water or gas production post perforation (b) Reservoir quality issue or high skin post perforation (c) Perforation effectiveness (d) Operation related issues (e) Post job issues like sanding or gas lift. Refer [Figure 19](#) for breakdown of these potential failure causes. High water cut or excessive gas production is a recurring challenge in brown fields due to the uncertainty in fluids contact movement overtime in some of the reservoirs. Following these findings, surveillance efforts and data acquisition like production logging, saturation logging (where operationally possible) has been enhanced to help mitigate such failures in future perforation jobs. Poor cement quality behind casing was also attributed to gas or water breakthrough in newly perforated intervals and hence requires additional measures during job planning and perforation interval selection. Reservoir quality and additional skin effects are mostly observed for new sands and in some cases where perforation gun selection has been identified as a potential issue. Perforation effectiveness and operation related issues are mostly related to gun selection and deployment which will be explained later in this section. Although perforation gun selection caused failure in only a few wells, it is also considered a key factor contributing to the lower than planned gains observed in many jobs as illustrated in [Figure 1](#). Another failure reason in some of the dual string wells was related to the magnetic orientation tool (MOT) which is used to orient the guns away from the sister string. To minimize such issues, it is recommended to conduct exhaustive system integrity check (SIT) and ensure availability of back-up and spares for the tool.
2. The success rate for cement packer perforation jobs was very low (<20%) as indicated in [Figure 20](#). The cement packer technique serves as an alternative to a full-scale workover to tap into the shallower opportunities above the existing production packer. Although considerably low cost than a workover, it remains a major expense operation and the high failure rate represents substantial value leakage for the operators. As shown in [Figure 2](#), cement packer jobs reported for only 10% of total perforation activities yet represented 42% of the total perforation costs, highlighting the extent of value leakage. The lookback study suggested several reasons for the failure of cement packer jobs such as poor-quality sands (Low Resistivity Low Contrast sands), operational issues during cementation jobs and perforation performance as documented by [Anand et al \(2024\)](#). While subsequent workflows and operational improvements have addressed the first two causes, a modified API 19B Sec II perforation testing is ongoing to suggest on the gun selection in this multi-tubular environment. Results from the testing will be published in a separate publication.

The additional testing requirement and poor perforation performance across multi-tubulars has been well documented by Melvyn et al (2009), Grove et al (2018), Burky et al (2018) and other authors.

3. The success rate of perforation jobs done with gun size smaller than 2" was very low. Majority of the well completion across various of the operator's fields comprise dual strings of size 2-7/8" or 2-3/8". The configuration is designed to ensure that most reservoirs in the stacked system are completed with selectivity. However, the small completion size prevents the deployment of through-tubing guns larger than 2". It was noted from the lookback study that the success rate for jobs completed with smaller guns (2" and below) was lower compared to larger sizes ($\geq 2\text{-}3/8"$) with 1-9/16" guns performing quite poorly (<20% success) as indicated in [Figure 21](#).

It should be noted that almost all gun selection for through tubing perforation jobs is done using commercial simulators which are primarily based on API 19B Sec I data (although the use of Sec II data to refine the outputs from the simulator has been growing in recent years). These simulators generally tend to provide optimistic results and often incorporate significant error as explained by Rajani et al (2017), Gladkikh et al (2009) etc which prompt the operator to select the smaller size. Based on the findings of this lookback study, the operators are limiting the use of guns smaller than 2" and exploring other options to perforate in the small completion sizes.

4. It was observed from the lookback study that in some jobs the gun type / size had to be changed during the execution, differing from what was originally planned in the job design as outlined in [Figure 22](#). This change was primarily due to logistical or operational reasons.

Although these changes did not directly result in significant number of failures, such adjustments are generally undesirable. Gun selection is highly specific to meet job objectives and any change from the original design requires a full re-evaluation. This is particularly critical for through-tubing perforation jobs, where performance is highly sensitive to gun size selection as explained in earlier in this section. It should be noted that the planned gun specifications were not properly documented for many of the jobs, which limited the ability to perform a more detailed analysis on the impact of changes in gun size selection.

5. The lookback also studies any potential trends between gun supplier and job success. [Figure 23](#) presents the breakdown of jobs across the different perforation gun suppliers engaged by the operators. This data suggests that there is no significant difference in perforation performance across the various gun suppliers. It should be noted, however, that the success or failure of a perforation job depends on multiple factors. Therefore, a larger dataset is required before drawing any conclusions on gun supplier performance. The operators are continuously updating the lookback dataset and it is expected that soon more conclusive insights will be drawn on gun supplier performance, particularly with respect to small size guns.

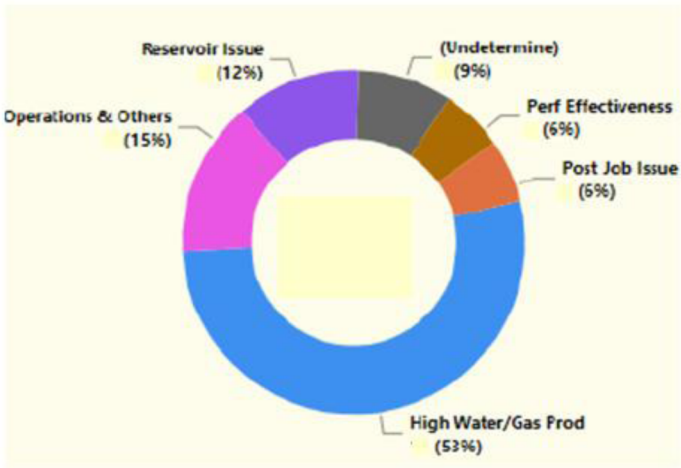


Figure 19—Main Reasons for Through Tubing Perforation Jobs Failure

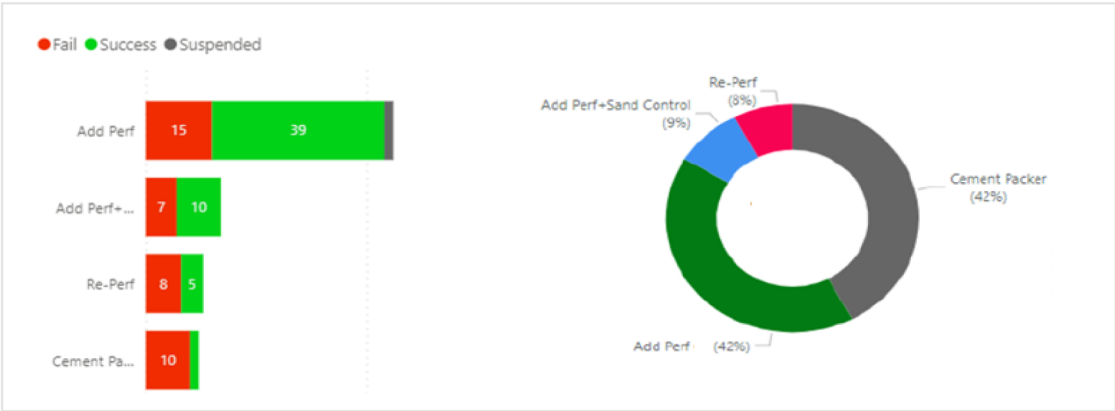


Figure 20—Cement Packer Low Success Rate (left) and Associated Value Leakages (right)

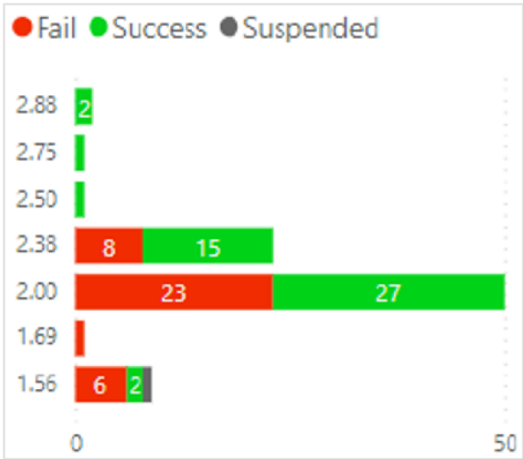


Figure 21—Poor Success Rate for Smaller Gun Sizes

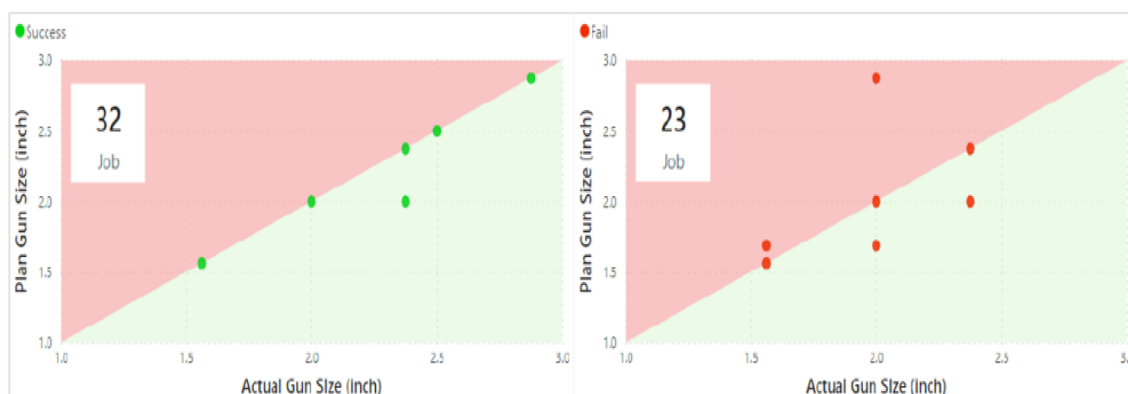


Figure 22—Planned and Actual Gun Used for Through Tubing Perforation Jobs

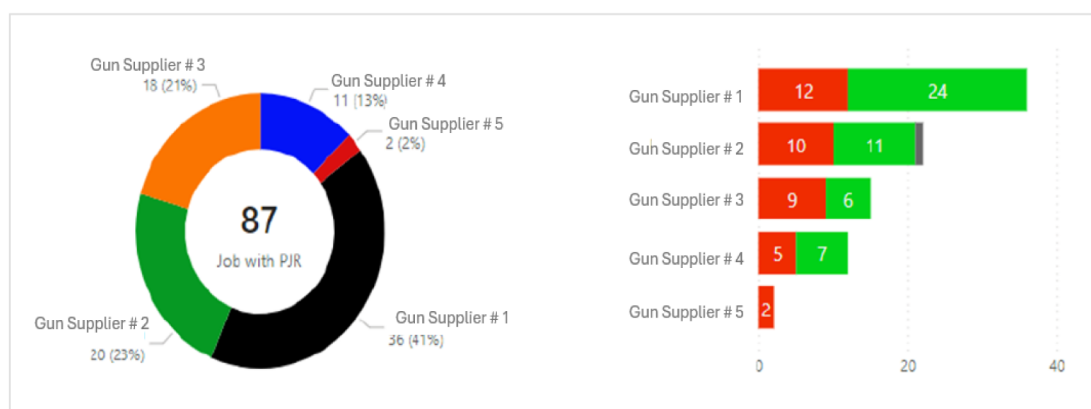


Figure 23—Relative Performance of Different Gun Suppliers for The Operators

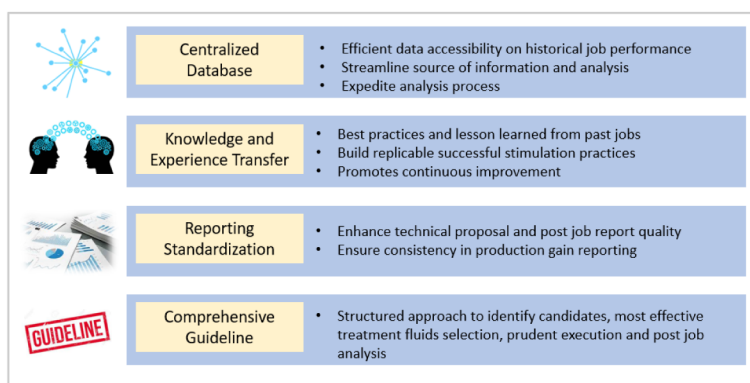


Figure 24—Value Creations from Stimulation and Add Perforation Lookback & Deep Dive Analysis

Summary of Recommendations from The Deep Dive Analysis for Historical Additional Perforation

Many findings were documented from the lookback study on additional perforation with some key insights highlighted above. These findings led to several key recommendations, which are crucial for minimizing similar failures in the future and achieving the target 80% success rate for additional perforation. The recommendations are summarized as follows:

1. Develop a thorough understanding on the Water Oil Contact (WOC) and Gas Oil Contact (GOC) prior to perforating any interval by reviewing available logs, maps and well test trend of nearby wells

producing from same reservoir. If there is still doubt about fluid contacts, it is highly recommended to conduct saturation logging.

2. Cement integrity is essential for zonal isolation and must be fully considered when finalizing the target perforation interval.
3. The team is recommended to fully comply to LRLC workflow developed within the operators when targeting an LRLC sand (Tham et al., 2020).
4. Conduct all required SIT and function tests for gun orientation systems and ensure suitable backup tools and spare parts are available on site.
5. Ensure that all jobs use the largest OD guns with the most effective and deep-penetrating charges. Guns smaller than 2" are not recommended, especially for multi-tubular applications.
6. Modified API 19B Sec II testing is recommended for gun selection in all cement packer or other complex completions.
7. Gun selection done during the planning stage and job design should not be changed during operation stage without conducting full necessary review.

Value Creations from Stimulation & Additional Perforation Lookback and Deep Dive Analysis

The lookback and deep dive analyses for stimulation and additional perforation have delivered clear and impactful value by driving tangible improvements in post-study. Additional perforation success rate rises from 57% to 80% and stimulation success rate dramatically increases from 54% to 83%. To further safeguard proposal quality and alignment with host authority requirements, MPM enforces the use of standardized techno-commercial proposal templates for stimulation and additional perforation activities, ensuring all submissions are comprehensive, technically sound and meet established technical criteria before approval. Complementing this, the establishment of a centralized Power BI database improves efficient access by the operators to Malaysia historical job performance data, streamlines information sourcing and analysis, and accelerates the technical evaluation process, enabling faster, more informed decision-making across all stages of project planning and execution. Through collaboration-driven learnings and capability building, these initiatives have fostered a culture of continuous improvement and knowledge sharing among stakeholders. Figure 24 summarizes the value propositions realized from this lookback and deep dive analysis. Crucially, these efforts have translated into significant cash values, minimized value leakage and safeguarding Malaysia's national production deliverables and reinforcing the benefits of a disciplined, data-informed approach. By providing operators with comprehensive, end-to-end guidelines grounded in robust field data, MPM empowers the operators to optimize well intervention activities, improve production delivery and ensure the sustainable long-term value of the nation's upstream assets within a competitive global landscape.

Conclusion and Way Forward

Building on the success of lookback and deep dive analyses in stimulation and additional perforation, the next phase focuses on scaling these lessons across other PE/IWR activities. By replicating the lookback approach across additional domains, MPM aims to institutionalize data-driven decision-making and continuous improvement throughout the asset lifecycle. Integration of advanced technologies such as artificial intelligence (AI) and digital diagnostics will further enhance the depth and speed of data analysis, enabling more precise damage diagnosis and well intervention planning. This technological advancement supports the development of pan-Malaysia benchmarks and real-time performance dashboards, providing operators and stakeholders with transparent, actionable insights to optimize asset performance consistently. Ultimately, these initiatives strengthen MPM's commitment as the host authority to safeguard national production, reduce value leakage and promote operational excellence across Malaysia's upstream sector.

The adoption of standardized methodologies, collaborative knowledge sharing, and digital innovation will continue to drive sustainable production gains and long-term value creation for the nation.

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